

JAVA ANIMAL INHERITANCE PROGRAM

Object-Oriented Programming Practical Record – Week 03

1. Aim

To write a Java program using inheritance and method overriding to display the name, habitat, food and sound of different animals.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named Week03.java inside the package packagecodejava.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create objects for Horse, Pig and Cow.
6. Each class inherits the properties and methods of Animal.
7. The sound() method is overridden in each animal class.
8. Display the animal details and sound.

3. Algorithm

1. Start the program.
2. Create the Animal parent class with name, habitat and food.
3. Create Cow, Pig and Horse classes extending Animal.
4. Override the sound() method in each child class.
5. Create objects for Horse, Pig and Cow in main().
6. Call show() for each object.
7. Display the animal name, habitat, food and sound.
8. Stop the program.

4. Explanation of the Program

Animal class: It is the parent class containing private data members name, habitat and food. It also contains getters, sound() and show() methods.

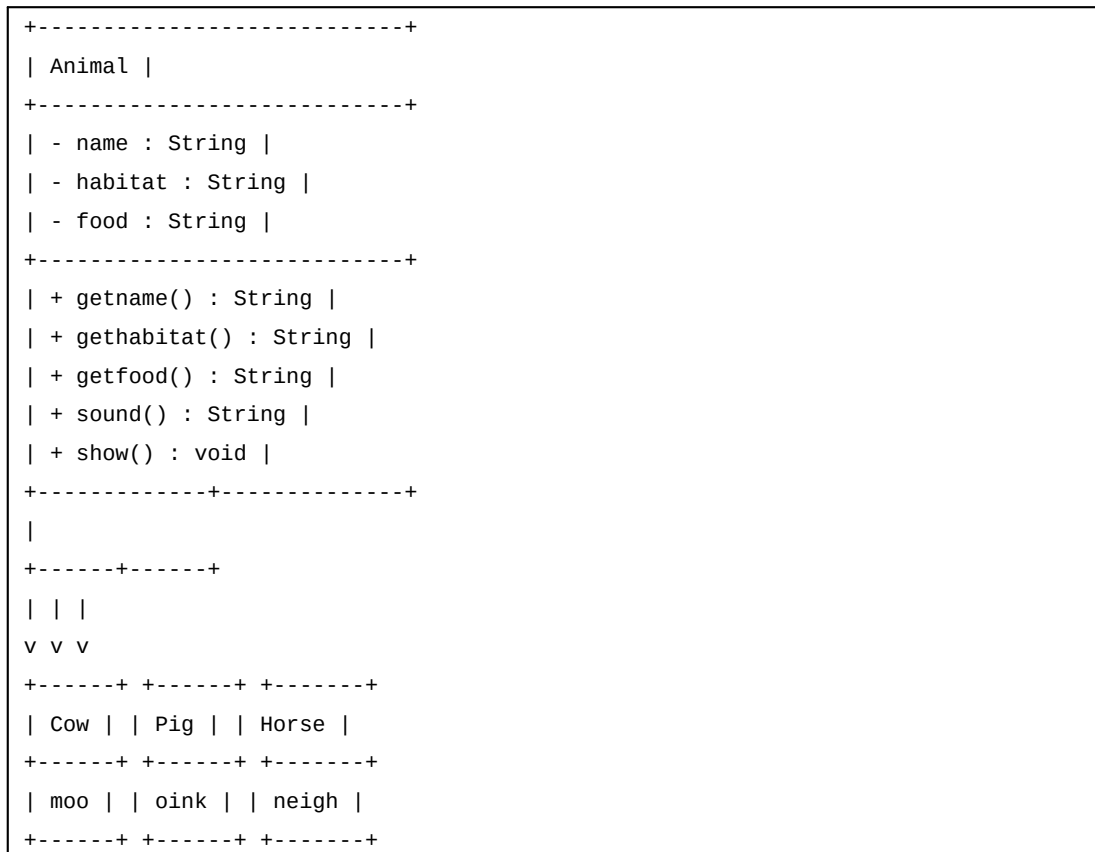
Cow class: It extends Animal and overrides sound() to return "moo".

Pig class: It extends Animal and overrides sound() to return "oink".

Horse class: It extends Animal and overrides sound() to return "neigh".

Method Overriding: Each child class provides its own implementation of the sound() method.

5. UML Class Diagram



6. Java Source Code

```
package packagecodejava;

class Animal {
    private String name;
    private String habitat;
    private String food;

    public Animal(String name, String habitat, String food) {
        this.name = name;
        this.habitat = habitat;
        this.food = food;
    }

    public String getname() {
        return name;
    }

    public String gethabitat() {
        return habitat;
    }

    public String getfood() {
        return food;
    }

    String sound() {
        return "animal sound";
    }

    void show() {
        System.out.println("animal is " + name);
        System.out.println("habitat is " + habitat);
        System.out.println("food is " + food);
        System.out.println("sound is " + sound());
        System.out.println("-----");
    }
}

class cow extends Animal {
    public cow() {
        super("cow", "shed", "grass");
    }

    @Override
    String sound() {
        return "moo";
    }
}

class pig extends Animal {
    public pig() {
        super("pig", "sty", "grains,vegetable");
    }

    @Override
    String sound() {
        return "oink";
    }
}

class horse extends Animal {
    public horse() {
        super("horse", "stable", "grass");
    }

    @Override
    String sound() {
```

```
        return "neigh";
    }
}

public class Week03 {
    public static void main(String[] args) {
        horse farm = new horse();
        farm.sound();
        farm.show();

        pig far = new pig();
        farm.sound();
        far.show();

        cow fa = new cow();
        fa.sound();
        fa.show();
    }
}
```

7. Output of the Program

```
animal is horse
habitat is stable
food is grass
sound is neigh
-----
animal is pig
habitat is sty
food is grains,vegetable
sound is oink
-----
animal is cow
habitat is shed
food is grass
sound is moo
-----
```

8. Result

Thus, the Java program successfully demonstrates inheritance and method overriding by displaying different sounds for Horse, Pig and Cow.